Recall:

1. Laplace Transform:

$$X(s) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-st} \, dt \quad , \quad s \in \mathbb{C}.$$

ROC:

is the collection of all s such that

$$\int_{-\infty}^{+\infty} \left| x(t) \cdot e^{-st} \right| dt < +\infty$$

2. Any vertical line within the ROC contains enough information to invert the Laplace transform.

If the line $Re(s) = \sigma$ is in the ROC, then:

$$x(t) = e^{\sigma t} \cdot \frac{1}{2\pi} \int_{-\infty}^{+\infty} X(\sigma + j\omega) \cdot e^{j\omega t} \, d\omega.$$

3. $u(t) \overset{L}{\longleftrightarrow} \frac{1}{s}$, $Re(s) > 0$

$-u(-t) \overset{L}{\longleftrightarrow} \frac{1}{s}$, $Re(s) < 0$

Ex: If we know that $X(s) = \frac{1}{s}$ for all s with

$1.5 \leq Re(s) \leq 2.3$

In this case, $x(t) = u(t)$.

I. Examples. (continued)

③ $x(t) = e^{-at}u(t)$, a is any complex number.

ROC:

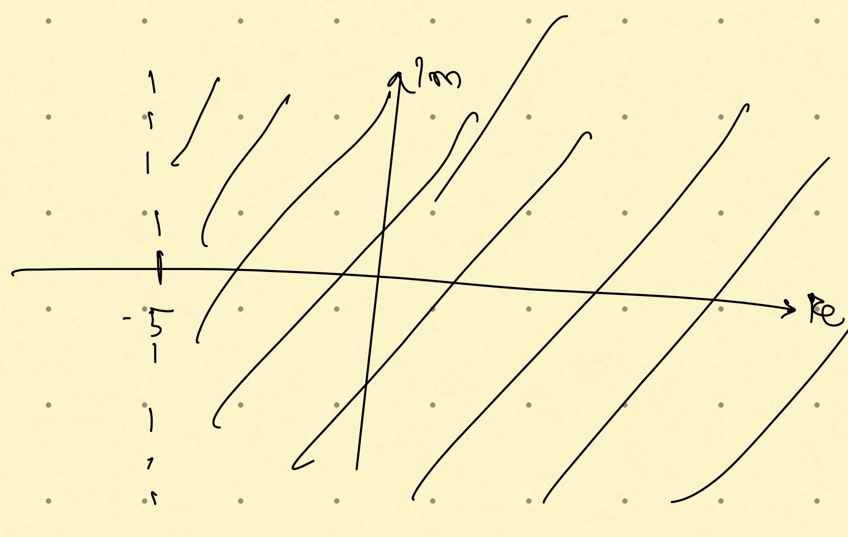
$$\int_{-\infty}^{+\infty} |x(t)\cdot e^{-st}|\,dt$$

$$= \int_0^{\infty} |e^{-at}\cdot e^{-st}|\,dt = \int_0^{\infty} |e^{-(s+a)t}|\,dt.$$

$$= \int_0^{\infty} e^{-\mathrm{Re}(s+a)t}\,dt.$$

$< +\infty$ iff: $\mathrm{Re}(s+a) > 0$

i.e. $\mathrm{Re}(s) > -\mathrm{Re}(a)$.



$a = 5+3j$

$\mathrm{Re}(a) = 5$

X(s): Assume $s \in$ ROC i.e. $\mathrm{Re}(s+a) > 0$.

$$X(s) = \int_0^{\infty} e^{-at}\cdot e^{-st}\,dt = \int_0^{\infty} e^{-(s+a)t}\,dt.$$

$$= \frac{1}{s+a}.$$

$$\therefore \quad e^{-at}u(t) \xleftrightarrow{L} \frac{1}{s+a}, \quad Re(s) > -Re(a).$$

(4) <u>Homework:</u>

$$-e^{-at}u(-t) \xleftrightarrow{L} \frac{1}{s+a}, \quad Re(s) < -Re(a).$$

<u>Aside:</u>

$$u(t) \xleftrightarrow{L} \frac{1}{s}, \quad Re(s) > 0$$

$$\downarrow$$

$$e^{-at}u(t) \xleftrightarrow{L} \frac{1}{s+a}, \quad Re(s+a) > 0.$$

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(5)
$$x(t) = \underbrace{3 \cdot e^{-2t}u(t)}_{x_1(t)} - \underbrace{2 \cdot e^{-t}u(t)}_{x_2(t)}.$$

<u>ROC</u> :

$$x(t) = x_1(t) + x_2(t)$$

$$\int_{-\infty}^{+\infty} |x(t)e^{-st}|\,dt = \int_{-\infty}^{+\infty} |x_1(t)e^{-st} + x_2(t)e^{-st}|\,dt.$$

$$\leq \int_{-\infty}^{+\infty} |x_1(t)e^{-st}|\,dt + \int_{-\infty}^{+\infty} |x_2(t)e^{-st}|\,dt.$$

$\Rightarrow$ If $s \in$ ROC of x_1 and $s \in$ ROC of x_2 then

$s \in$ ROC of $(x_1 + x_2)$.

i.e. ROC $(x_1 + x_2)$ contains (ROC of x_1) $\cap$ (ROC of x_2).

Apply to the current example:

ROC of $x(t)$ contains $(Re(s) > -2) \cap (Re(s) > -1)$

i.e. ROC of $x(t)$ contains $Re(s) > -1$.

We need to check if $Re(s) > -1$ is the entire ROC.

To do so, pick any s with $Re(s) \leq -1 \rightarrow Re(s) = \sigma$.

$$\int_{-\infty}^{+\infty} |x(t) \cdot e^{-st}| dt = \int_{0}^{\infty} |(3 \cdot e^{-2t} - 2e^{-t}) e^{-st}| dt.$$

$$= \int_{0}^{\infty} |3 \cdot e^{-(\sigma+2)t} - 2 \cdot e^{-(\sigma+1)t}| dt.$$

Since $Re(s) \leq -1$ we have $\sigma + 1 \leq 0$

$$= \int_{0}^{\infty} |2e^{-(\sigma+1)t} - 3e^{-(\sigma+2)t}| dt.$$

$$= \int_{0}^{\infty} e^{-(\sigma+1)t} |2 - 3 \cdot e^{-t}| dt.$$

Note that $(\sigma+1)t \leq 0$ i.e. $e^{-(\sigma+1)t} \geq 1$.

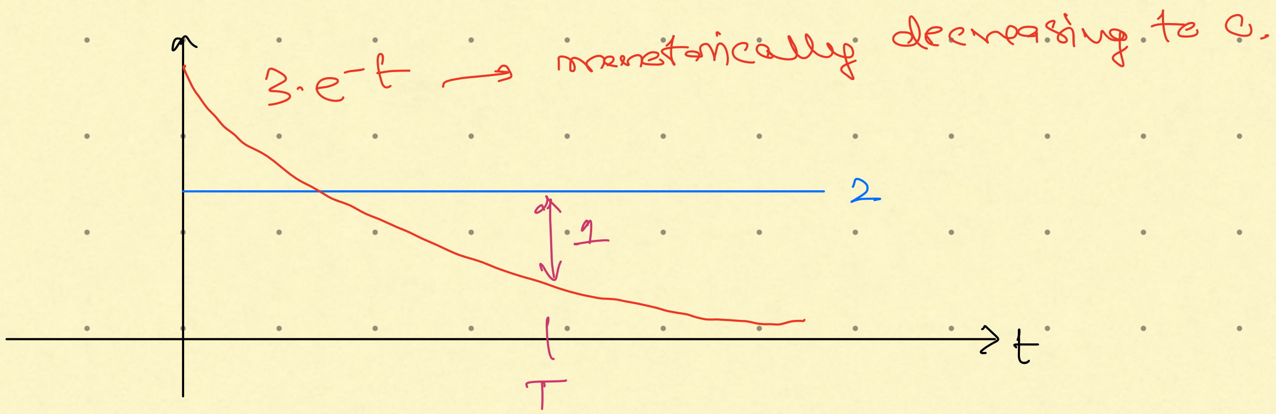
$$\geq \int_0^\infty 1 \cdot |2 - 3 \cdot e^{-t}| \, dt.$$

$$= \int_0^T |2 - 3e^{-t}| \, dt + \int_T^\infty |2 - 3e^{-t}| \, dt$$

$$\geq 0 + \int_T^\infty 1 \cdot dt \geq +\infty,$$

here we chose T such that for all $t \geq T$

$$|2 - 3 \cdot e^{-t}| \geq 1.$$



<u>Remark:</u> If $x = x_1 + x_2$ then:

ROC of x contains $(\text{ROC } x_1) \cap (\text{ROC } x_2)$ as a subset.

In general, we do not know if $(\text{ROC } x_1) \cap (\text{ROC } x_2)$ is equal to ROC of x.

- In the previous example:

$$(\text{ROC } x_1) \cap (\text{ROC of } x_2) = \text{ROC of } x.$$

- As another example, consider:

$$0 = e^{-t}u(t) + [-e^{-t}u(t)].$$

$$\text{ROC }(x_1 + x_2) = \text{ROC of } 0 = \text{entire s-plane}$$

$$\text{ROC}(x_1) = \text{ROC}(x_2) = \text{ROC}(x_1) \cap \text{ROC}(x_2)$$

$$= \text{Re}(s) > -1.$$

In this case:

$$\text{ROC of } x_1 \cap \text{ROC of } x_2 \not\subseteq \text{ROC of } x$$